

MongoDB Practice Task

1. Create a Database and Collection

- use library
- db.createCollection("books")

```
test> use library
switched to db library
library> db.createCollection("books")
{ ok: 1 }
library> db.createCollection("novel")
{ ok: 1 }
library> show collections
MongoshInvalidInputError: [COMMON-10001] 'collections' is not a valid argument for "show".
library> show collections
books
novel
library> db.novel.drop()
true
library> show collections
books
```

2. Insert Operations

- Insert Document

```
db.books.insertOne({
  title: "The Great Gatsby",
  author: "F.Scott Fitzgerald",
  published_year: 1925
})
```

- **Insert Multiple Document**

```
db.books.insertMany([
  { title: "Harry Potter and the Philosopher's Stone", author:
    "J.K.Rowling", published_year: "1997" },
  { title: "Wings of Fire", author: "A.P.J.Abdul Kalam",
    published_year: "1999" },
  { title: "Rich Dad Poor Dad", author: "Robert T.Kiyosaki",
    published_year: "1997" }
])
```

```
library> db.books.insertMany([
  { title: "The Great Gatsby", author: "F.Scott Fitzgerald", published_yea
r: 1925, price: 300 },
  { title: "Harry Potter", author: "J.K.Rowling", published_year: 1997, pr
ice: 500 },
  { title: "Wings of Fire", author: "A.P.J.Abdul Kalam", published_year: 1
999, price: 400 },
  { title: "Rich Dad Poor Dad", author: "Robert T.Kiyosaki", published_yea
r: 1997, price: 450 },
  { title: "Book A", author: "John", published_year: 2005, price: 300 },
  { title: "Book B", author: "John", published_year: 2010, price: 500 },
  { title: "Book C", author: "Sara", published_year: 2015, price: 400 },
  { title: "Book D", author: "Sara", published_year: 2018, price: 600 },
  { title: "Book E", author: "Ali", published_year: 2020, price: 700 }
])
```

3. Read Operations

- Display All books

```
db.books.find()
```

- Filter by condition

```
db.books.find({ published_year: "1997" })
```

```

]
library> db.books.find().sort({published_year:1})
[
  {
    _id: ObjectId('69f720d15f6d6b0aea3682d1'),
    title: 'The Great Gatsby',
    author: 'F.Scott Fitzgerald',
    published_year: 1925
  },
  {
    _id: ObjectId('69f722ec5f6d6b0aea3682d3'),
    title: 'The Alchemist',
    author: 'Paulo Coelho',
    published_year: '1988'
  },
  {
    _id: ObjectId('69f722ec5f6d6b0aea3682d2'),
    title: "Harry Potter and the Philosopher's Stone",
    author: 'J.K.Rowling',
    published_year: '1997'
  },
  {
    _id: ObjectId('69f722ec5f6d6b0aea3682d5'),
    title: 'Rich Dad Poor Dad',
    author: 'Robert T.Kiyosaki',
    published_year: '1997'
  },
  {
    _id: ObjectId('69f722ec5f6d6b0aea3682d4'),
    title: 'Wings of Fire',
    author: 'A.P.J.Abdul Kalam',
    published_year: '1999'
  }
]

```

4. Update Operations

- Update One Document

```

db.books.updateOne(
  { title: "The Alchemist" },
  { $set: { published_year: "1987" } }
)

```

```

library> db.books.updateOne({title:"The Alchemist"},{$set:{published_year:"1987"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
library> db.books.find({title:"The Alchemist"})
[
  {
    _id: ObjectId('69f722ec5f6d6b0aea3682d3'),
    title: 'The Alchemist',
    author: 'Paulo Coelho',
    published_year: '1987'
  }
]
library> db.books.deleteOne({title:"The Alchemist"})
{ acknowledged: true, deletedCount: 1 }
library> db.books.find()
[
  {
    _id: ObjectId('69f720d15f6d6b0aea3682d1'),
    title: 'The Great Gatsby',
    author: 'F.Scott Fitzgerald',
    published_year: 1925
  },
  {
    _id: ObjectId('69f722ec5f6d6b0aea3682d2'),
    title: 'Harry Potter and the Philosopher's Stone',
    author: 'J.K.Rowling',
    published_year: '1997'
  },
  {
    _id: ObjectId('69f722ec5f6d6b0aea3682d4'),
    title: 'Wings of Fire',
    author: 'A.P.J.Abdul Kalam',
    published_year: '1999'
  },
]

```

5. Delete Operation

```
db.books.deleteOne({ title: "The Alchemist" })
```

```

library> db.books.deleteOne({title:"The Alchemist"})
{ acknowledged: true, deletedCount: 1 }
library> db.books.find()
[
  {
    _id: ObjectId('69f720d15f6d6b0aea3682d1'),
    title: 'The Great Gatsby',
    author: 'F.Scott Fitzgerald',
    published_year: 1925
  },
  {
    _id: ObjectId('69f722ec5f6d6b0aea3682d2'),
    title: 'Harry Potter and the Philosopher's Stone',
    author: 'J.K.Rowling',
    published_year: '1997'
  },
  {
    _id: ObjectId('69f722ec5f6d6b0aea3682d4'),
    title: 'Wings of Fire',
    author: 'A.P.J.Abdul Kalam',
    published_year: '1999'
  },
  {
    _id: ObjectId('69f722ec5f6d6b0aea3682d5'),
    title: 'Rich Dad Poor Dad',
    author: 'Robert T.Kiyosaki',
    published_year: '1997'
  }
]

```

6. Query Operators

- Sorting Operations

a. Ascending Order

```
db.books.find().sort({ published_year: 1 })
```

```

library> db.books.find().sort({ published_year: 1 })
[
  {
    _id: ObjectId('69f738b85f6d6b0aea3682db'),
    title: 'The Great Gatsby',
    author: 'F.Scott Fitzgerald',
    published_year: 1925,
    price: 300
  },
  {
    _id: ObjectId('69f738b85f6d6b0aea3682dc'),
    title: 'Harry Potter',
    author: 'J.K.Rowling',
    published_year: 1997,
    price: 500
  },
  {
    _id: ObjectId('69f738b85f6d6b0aea3682de'),
    title: 'Rich Dad Poor Dad',
    author: 'Robert T.Kiyosaki',
    published_year: 1997,
    price: 450
  },
  {
    _id: ObjectId('69f738b85f6d6b0aea3682dd'),
    title: 'Wings of Fire',
    author: 'A.P.J.Abdul Kalam',
    published_year: 1999,
    price: 400
  },
  {
    _id: ObjectId('69f738b85f6d6b0aea3682df'),
    title: 'Book A',
    author: 'John',
    published_year: 2005,
    price: 300
  },
  {
    _id: ObjectId('69f738b85f6d6b0aea3682e0'),

```

b. Descending Order

```
db.books.find().sort({ published_year: -1 })
```

```

library> db.books.find().sort({ published_year: 1 })
]
library> db.books.find().sort({ published_year: -1 })
[
  {
    _id: ObjectId('69f738b85f6d6b0aea3682e3'),
    title: 'Book E',
    author: 'Ali',
    published_year: 2020,
    price: 700
  },
  {
    _id: ObjectId('69f738b85f6d6b0aea3682e2'),
    title: 'Book D',
    author: 'Sara',
    published_year: 2018,
    price: 600
  },
  {
    _id: ObjectId('69f738b85f6d6b0aea3682e1'),
    title: 'Book C',
    author: 'Sara',
    published_year: 2015,
    price: 400
  },
  {
    _id: ObjectId('69f738b85f6d6b0aea3682e0'),
    title: 'Book B',
    author: 'John',
    published_year: 2010,
    price: 500
  },
  {
    _id: ObjectId('69f738b85f6d6b0aea3682df'),
    title: 'Book A',
    author: 'John',
    published_year: 2005,
    price: 300
  },
  {
    _id: ObjectId('69f738b85f6d6b0aea3682dd'),

```

7. Aggregation Operations

- Filter Books after 2010

```
db.books.aggregate([ { $match: { published_year: { $gt: 2010 } } } ])
```

```

]
library> db.books.aggregate([
|   { $match: { published_year: { $gt: 2010 } } }
| ])
[
|   {
|     _id: ObjectId('69f738b85f6d6b0aea3682e1'),
|     title: 'Book C',
|     author: 'Sara',
|     published_year: 2015,
|     price: 400
|   },
|   {
|     _id: ObjectId('69f738b85f6d6b0aea3682e2'),
|     title: 'Book D',
|     author: 'Sara',
|     published_year: 2018,
|     price: 600
|   },
|   {
|     _id: ObjectId('69f738b85f6d6b0aea3682e3'),
|     title: 'Book E',
|     author: 'Ali',
|     published_year: 2020,
|     price: 700
|   }
| ]
library> db.books.aggregate([
|   { $sort: { published_year: -1 } }
| ])

```

- Sort Books by Latest Year

```
db.books.aggregate([ { $sort: { published_year: -1 } } ])
```

```

]
library> db.books.aggregate([
|   { $sort: { published_year: -1 } }
| ])
[
|   {
|     _id: ObjectId('69f738b85f6d6b0aea3682e3'),
|     title: 'Book E',
|     author: 'Ali',
|     published_year: 2020,
|     price: 700
|   },
|   {
|     _id: ObjectId('69f738b85f6d6b0aea3682e2'),
|     title: 'Book D',
|     author: 'Sara',
|     published_year: 2018,
|     price: 600
|   },
|   {
|     _id: ObjectId('69f738b85f6d6b0aea3682e1'),
|     title: 'Book C',
|     author: 'Sara',
|     published_year: 2015,
|     price: 400
|   },
|   {
|     _id: ObjectId('69f738b85f6d6b0aea3682e0'),
|     title: 'Book B',
|     author: 'John',
|     published_year: 2010,
|     price: 500
|   },
|   {
|     _id: ObjectId('69f738b85f6d6b0aea3682df'),
|     title: 'Book A',
|     author: 'John',
|     published_year: 2005,
|   }
| ]

```


- Top 3 Expensive Books

```
db.books.aggregate([ { $sort: { price: -1 } }, { $limit: 3 } ])
```

```
library> db.books.aggregate([
|   { $sort: { price: -1 } },
|   { $limit: 3 }
| ])
|
| [
|   {
|     _id: ObjectId('69f738b85f6d6b0aea3682e3'),
|     title: 'Book E',
|     author: 'Ali',
|     published_year: 2020,
|     price: 700
|   },
|   {
|     _id: ObjectId('69f738b85f6d6b0aea3682e2'),
|     title: 'Book D',
|     author: 'Sara',
|     published_year: 2018,
|     price: 600
|   },
|   {
|     _id: ObjectId('69f738b85f6d6b0aea3682e0'),
|     title: 'Book B',
|     author: 'John',
|     published_year: 2010,
|     price: 500
|   }
| ]
library> db.books.aggregate([
```

- Counts Books per Author

```
db.books.aggregate([ { $group: { _id: "$author",
                                totalBooks: { $sum: 1 } } } ])
```

```
library> db.books.aggregate([
|   {
|     $group: {
|       _id: "$author",
|       totalBooks: { $sum: 1 }
|     }
|   }
| ])
|
| [
|   { _id: 'John', totalBooks: 2 },
|   { _id: 'J.K.Rowling', totalBooks: 1 },
|   { _id: 'F.Scott Fitzgerald', totalBooks: 1 },
|   { _id: 'A.P.J.Abdul Kalam', totalBooks: 1 },
|   { _id: 'Sara', totalBooks: 2 },
|   { _id: 'Ali', totalBooks: 1 },
|   { _id: 'Robert T.Kiyosaki', totalBooks: 1 }
| ]
library> db.books.aggregate([
|   {
|     $group: {
```

- Average Price per Author

```
db.books.aggregate([ { $group: { _id: "$author",
avgPrice: { $avg: "$price" } } } ])
```

```
library> db.books.aggregate([
  {
    $group: {
      _id: "$author",
      avgPrice: { $avg: "$price" }
    }
  }
])
[
  { _id: 'Robert T.Kiyosaki', avgPrice: 450 },
  { _id: 'J.K.Rowling', avgPrice: 500 },
  { _id: 'F.Scott Fitzgerald', avgPrice: 300 },
  { _id: 'A.P.J.Abdul Kalam', avgPrice: 400 },
  { _id: 'Sara', avgPrice: 500 },
  { _id: 'Ali', avgPrice: 700 },
  { _id: 'John', avgPrice: 400 }
]
```

- Highest Price per Author

```
db.books.aggregate([ { $sort: { price: -1 } }, { $group: {
_id: "$author", highestPrice: { $first: "$price" } } } ])
```

```
library> db.books.aggregate([
  { $sort: { price: -1 } },
  {
    $group: {
      _id: "$author",
      highestPrice: { $first: "$price" }
    }
  }
])
[
  { _id: 'F.Scott Fitzgerald', highestPrice: 300 },
  { _id: 'J.K.Rowling', highestPrice: 500 },
  { _id: 'Ali', highestPrice: 700 },
  { _id: 'Sara', highestPrice: 600 },
  { _id: 'Robert T.Kiyosaki', highestPrice: 450 },
  { _id: 'A.P.J.Abdul Kalam', highestPrice: 400 },
  { _id: 'John', highestPrice: 500 }
]
```

- Total Number of Books

```
db.books.aggregate([ { $group: { _id: null, totalBooks:
{ $sum: 1 } } } ])
```

```
{ _id: 'Robert T. Kiyosaki', highestPrice: 450 },
{ _id: 'A.P.J. Abdul Kalam', highestPrice: 400 },
{ _id: 'John', highestPrice: 500 }
]
library> db.books.aggregate([
|   {
|     $group: {
|       _id: null,
|       totalBooks: { $sum: 1 }
|     }
|   }
| ])
[ { _id: null, totalBooks: 9 } ]
library>
```